Vanshaj Verma

Roll No - 2023UIC4121
Bachelor of Technology – Instrumentation and Control
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PROFESSIONAL SUMMARY

Data Analyst with experience in **data pipelining, explainable modeling, and data quality validation**. Semi-Finalist, **American Express** Product Analytics Track. Eliminated a **19.7% silent error rate** in HRIS imports at Diversio and improved data consistency **125x** at CSIR-NPL using **Python, SQL, and FastAPI**.

EDUCATION

Netaji Subhas University of Technology (NSUT), Delhi — **Aug 2023 – May 2027**
Bachelor of Technology (B.Tech) – Instrumentation and Control Engineering

WORK EXPERIENCE

- **Diversio Ltd.** – *SDE Intern* | **Project** — *Remote*
  HRIS Import Validator & Org-Health Analytics (Python, FastAPI, pytest, GitHub Actions) — *Aug 2026 – Present*
  - **Data Validation & Benchmarking:** Benchmarked an HRIS validator ($O(n)$ cycle detection) on 150k rows, achieving 100% targeted error detection with zero false positives.
  - **Explainable Modeling:** Designed a fix-suggestion model (99.4% precision, 90.2% coverage); used department signals to cut incorrect departed-manager suggestions by 3.8x.
  - **Data Quality & Optimization:** Deployed a review queue yielding a 3-point data quality bump via one-click fixes; eliminated a 19.7% silent error rate by patching sequential ID flaws.

- **CSIR-National Physical Laboratory** – *Research & Development Intern* — *Delhi, India*
  Instrument Automation & Pipeline Development (Python, PyVISA, SCPI) — *May 2026 – Jul 2026*
  - **Data Ingestion & Automation:** Built Python GUI software to collect and structure lab measurements, replacing a manual LabVIEW workflow.
  - **Data Validation & Logging:** Added error handling and activity logging to catch failures early, ensuring clean records and exact-source debugging.
  - **Pipeline Engineering & Analytics:** Processed 134k measurements across 57 sessions; improved consistency 125x via a quality pipeline, surfacing a strong correlation ($r = 0.959$).

PROJECTS

- **Parakh: does a stock signal beat the NIFTY 50?** | *Python, SQL, PostgreSQL, LightGBM* — **GitHub**
  - **Data Pipeline & Benchmarking:** Automated a 10-year PostgreSQL price pipeline with raw feed validation; pre-registered a holdout where a simple rule beat LightGBM (t=2.39 vs 1.65).
  - **SQL Analytics & Insight:** Analyzed 38k backtests in SQL; proved models couldn't beat a coin-flip (48.7% accuracy) and the ∼0.2% algorithmic edge was smaller than trading costs.

- **Delhi Air Quality Analysis** | *SQL, Python, FastAPI, Plotly, MapLibre* — **GitHub**
  - **Data Pipeline & Quality:** Unified 106k station-days across 3 disparate sources into SQLite (r = 0.96–0.97 vs official records), catching and filtering critical PM2.5 and CO sensor errors.
  - **Statistical Insight:** Proved winter smog is mostly weather-driven (emissions only 9-17% above baseline); isolated a 4-week crop-burning window running 2.2x above baseline.

- **TalentScout: AI Hiring Assistant** | *Python, PostgreSQL, Groq LLM, Docker* — **GitHub**
  - **Evaluation Pipeline:** Built an LLM data parsing pipeline; designed an evaluation harness against labeled cases achieving 100% extraction accuracy and 97% skill-matching F1.
  - **Explainability & Bias Testing:** Developed a rubric-based AI grader with prompt-injection detection; conducted bias testing that halved the fluent vs. non-native English scoring gap.

TECHNICAL SKILLS

- **Languages & Tools**: Python, SQL, C++, Pandas, NumPy, Plotly, FastAPI, Excel, LLM APIs, Git, Docker, GitHub Actions
- **Analytics**: A/B Testing, Metric Design, Data Pipeline Engineering, Statistical Benchmarking

ACHIEVEMENTS

- **Semi-Finalist, American Express Product Analytics Track:** Analyzed clickstream and behavioral data in Python and Pandas, answering 6 business questions on conversion and segmentation.
- **EY Techathon 6.0 (Advanced to Round 2 – BFSI, Tata Capital):** Defined metrics and created synthetic datasets for a 4-member team building an agentic AI loan-sales chatbot via LangGraph.